

1. *Team Information*

Names (UNIs): Adi Gal (aeg2225), Alexandra Gao (atg2152), Je Seung (Jason) You (jy2908), Swan Htet (sh3969)

Team Number: 26

Project Name: "Media Map"

2. *Prior Individual Project*

We are expanding on Adi's individual project. We are reassessing the target user population by focusing on movies and TV shows exclusively (rather than books, movies, and TV shows), and we will expand on the project by doing more research with our users and by creating a basic implementation.

3. *Target User Population*

Our target user population is made of people in the teen to older adult age range who enjoy watching movies and TV shows and who want recommendations.

4. *Project concept*

This project tackles a subproblem of a larger issue.

The larger problem faced by users is effective media-finding: users want to get recommendations for film and television titles from algorithms, determine whether they want to watch it, and then access the shows.

Among other inefficiencies, the recommendation process is opaque. Users cannot understand, visualize, or control why a website's recommendation algorithm is recommending a particular show, even if that would help them find the show they want. While other problems in the media-finding process are handled by other websites, like Netflix, IMDB, and Goodreads, there is no available source which allows users to clarify and control the recommendation process. Focusing on this subproblem lets our project serve an original and currently-unmet user need.

The design solution is fundamentally to create "links," or connections, between different film and television titles. Our site will use these connections as a source of recommendations. That means that media titles will be recommended on the basis of criteria like 'same author,' 'same director,' 'similar story,' and so on. By recommending movies and TV shows on the basis of specific, named connections, we allow users to understand why they are being recommended a particular title. Users will also be able to choose which links they want to prioritize. This will allow them to customize why they are receiving particular recommendations.

5. *Challenge*

The project has a broad scope, and is a significant creative challenge.

The problems users face—understanding and controlling their movie and television recommendations—is extremely broad. In order to give users the ability to control their own recommendation process, our design solution has to acknowledge many different priorities from users of all different kinds. It also has to synthesize various different aspects of a movie or TV show in order to find similarities between different titles which exist for different reasons.

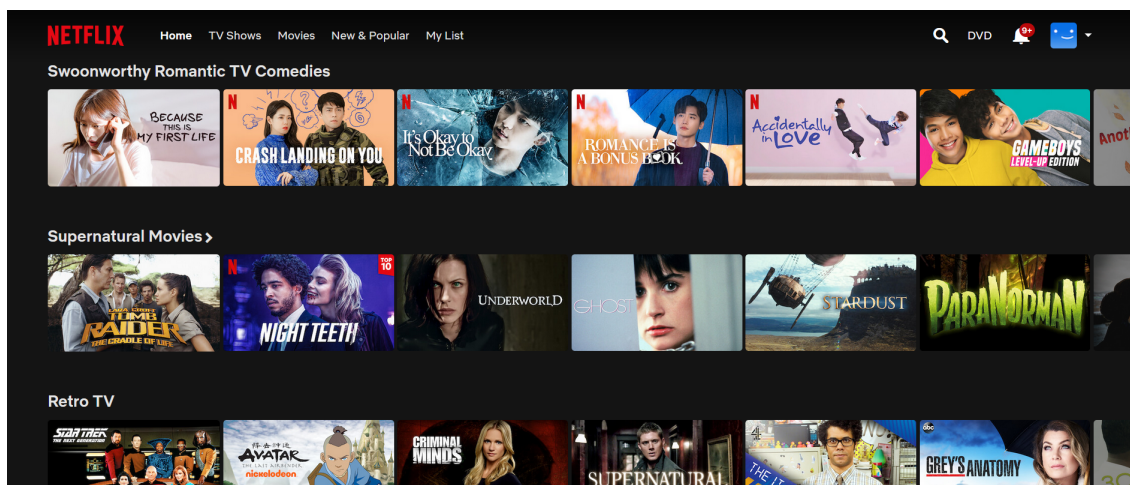
The project is also a creative design challenge. We are reimagining the traditional model of movie and TV recommendations in a way which must be innovative enough to engage users, intuitive enough to be usable, and which will give users understanding and control of the recommendation process. Figuring out how to let users best control their recommendations is a significant challenge.

6. *Comparative analysis*

Many popular streaming services (Netflix, Hulu, Disney+, Peacock) use what we call the ‘Blockbusters’ model for recommendations. They show a horizontal array of movie and TV shows covers, similar to a shelf at the defunct video rental service Blockbuster, and each row is recommended based on a criterion chosen by the site, not the user.

These services do well by combining their central service—providing users with movies and TV that they can watch online—with recommendations. This creates a hub where users can find a show and immediately start watching it. They also do well by making the design similar to real-world designs—like shelves at a video store.

However, where they fail is in their lack of explanation and control. Users cannot choose how movies and TV shows are recommended to them, and they do not understand why these titles are recommended. Also, there is no way to get recommendations between streaming services—each service recommends only the titles that it owns, even if there are titles which the user would like better.

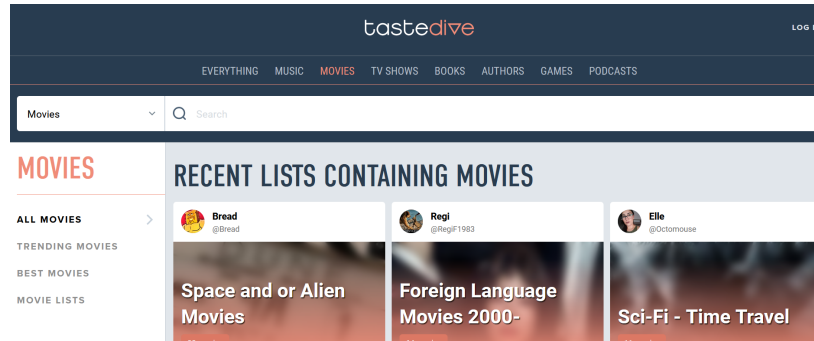


A page from the popular streaming service Netflix showing horizontal rows of visual media. The rows are labeled ‘swoon worthy romantic TV comedies,’ ‘supernatural movies,’ and ‘retro TV.’ There is no way to choose what options are shown.

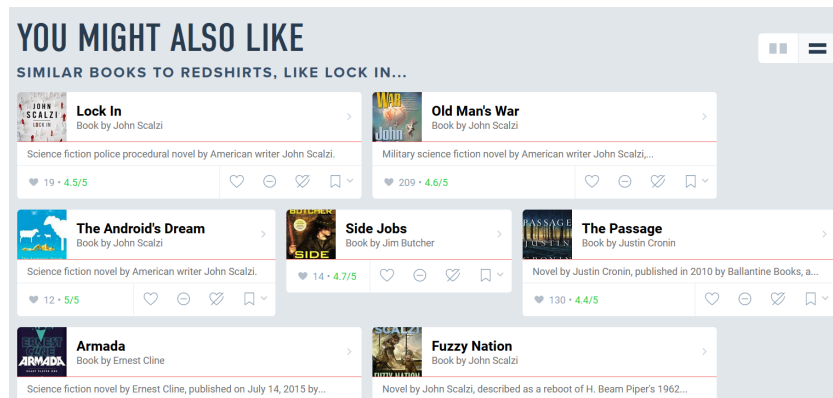
There are also less-popular sites which are dedicated to finding new movies and TV shows, like TasteDive. TasteDive shows lists curated by individuals collecting different movies, TV shows, books, authors, games, podcasts, or music. It also shows which media titles of each type are ‘trending,’ and which are the ‘best.’ Clicking on an individual media title provides you with a brief summary of the story, links to streaming services where you can watch the title, and a number of media titles which are “similar” to it. No explanation is provided about how this similarity is determined.

TasteDive succeeds by collecting many different types of media in one place. It allows users to see media titles from a number of streaming services, not just one, and gives users access to them (by showing where users can stream that title). It also allows users to create lists and view the lists of other users. These are important efficiencies.

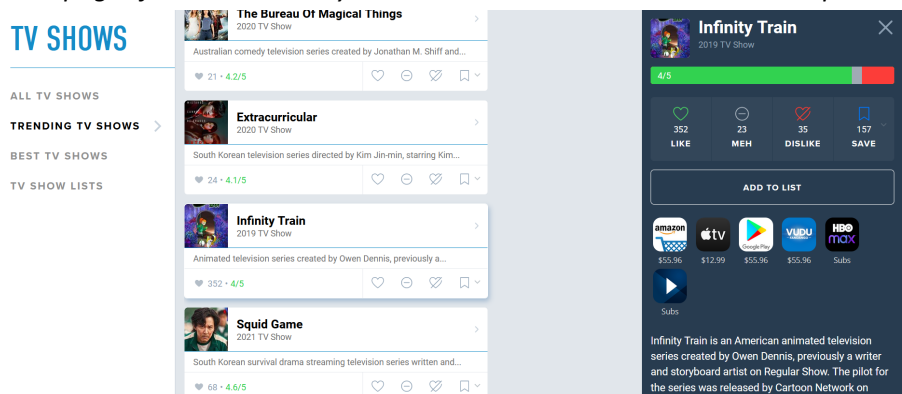
However, like the streaming services, TasteDive does not offer users full control or understanding of their recommendations. The only actual recommendations on the site are based on the undefined characteristics of “trending,” “best,” or “similar,” or through the individually-curated lists.



TasteDive's front page takes you to different types of media: 'everything,' music, movies, TV shows, books, authors, games, and podcasts. On the 'Movies' page, we see user-curated lists containing movies, trending movies, and 'best' movies.



Individual pages for a book show you books which are "similar" without explaining why.

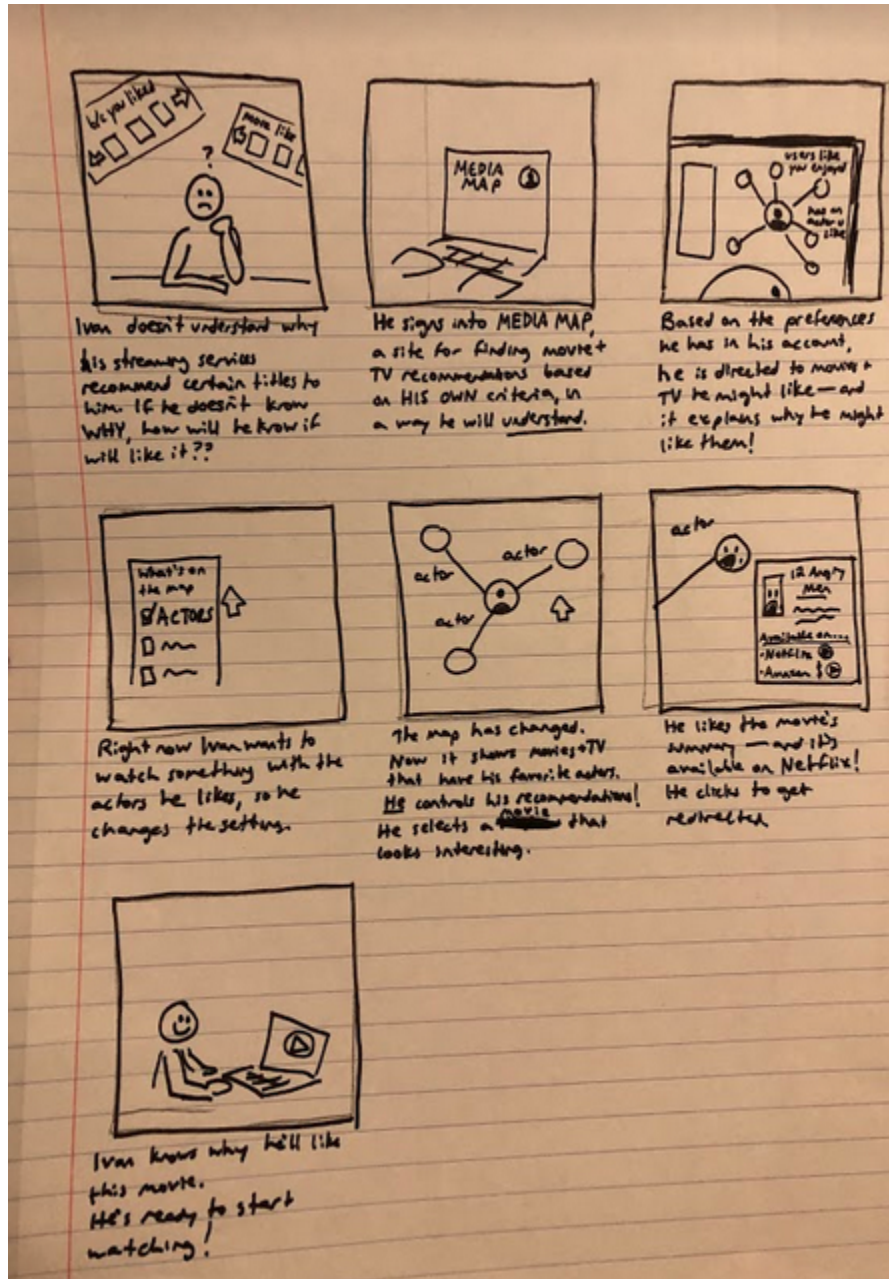


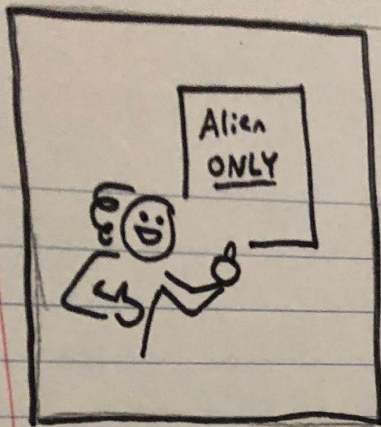
TasteDive provides a summary of the media title, the ability to add it to a list, as well as links to where it is available to stream (and for how much money.)

7. Explanation of design solution

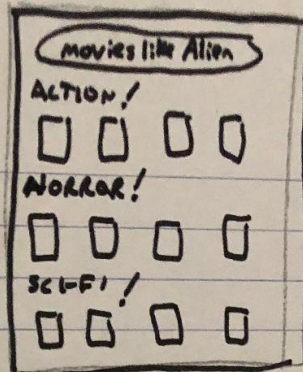
As explained in our project concept, our design solution is to provide more control on the user's end in terms of personalizing the recommendation algorithm of movies and TV shows. In this regard, our design solution is to create a platform that grants users freedom to choose/provide their "key attribute" that is applied towards our recommendation algorithm. This personalized recommendation algorithm will be generated by Relational database or GraphQL database, and we will implement this algorithm to our web application.

8. Storyboard(s)

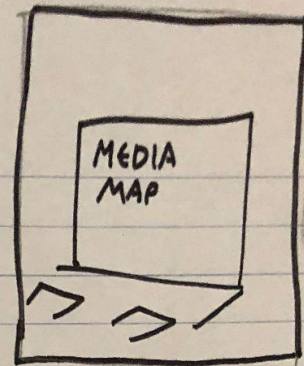




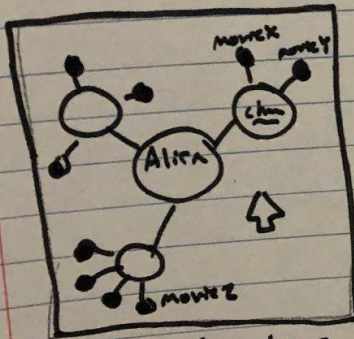
Nina is a big fan of the movie *Alien*, but only the characters — she normally hates that genre.



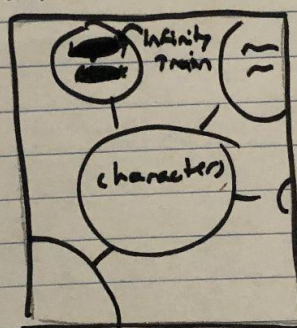
But traditional searches only give her movies + TV similar to *Alien*'s genre — not the characters! She can't control the recommendation to look for what she wants.



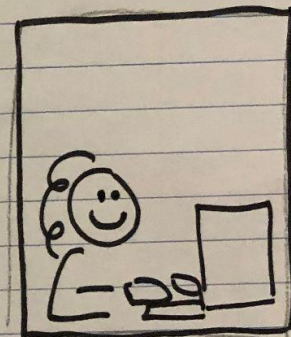
So she goes to Media Map, a site for controlling your recommendations.



Media Map shows her a map of other movies + TV shows with similar attributes to *Alien*.

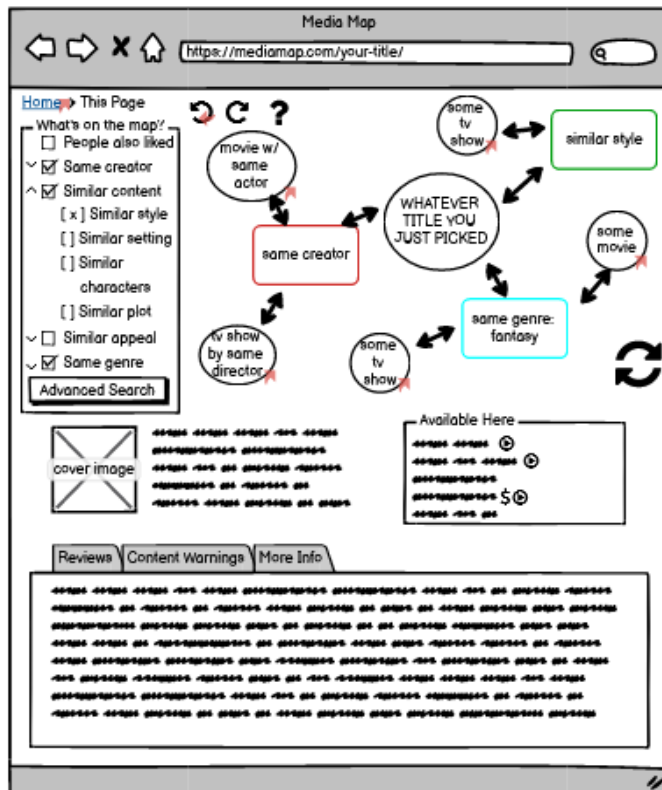
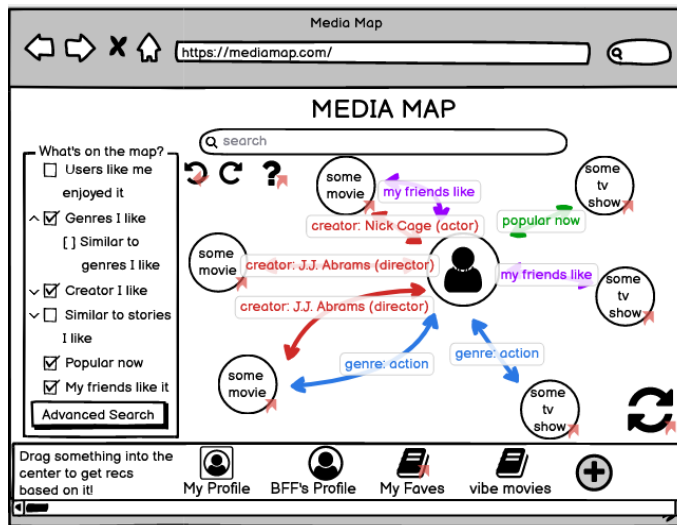


Look! Here are some movies + shows with similar characters, which aren't necessarily in the same genre!

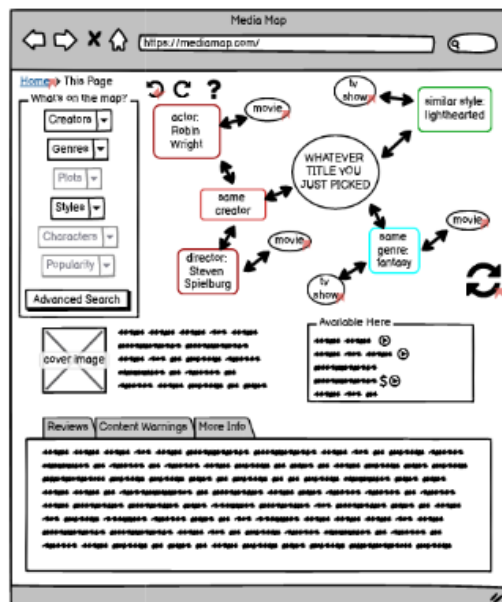
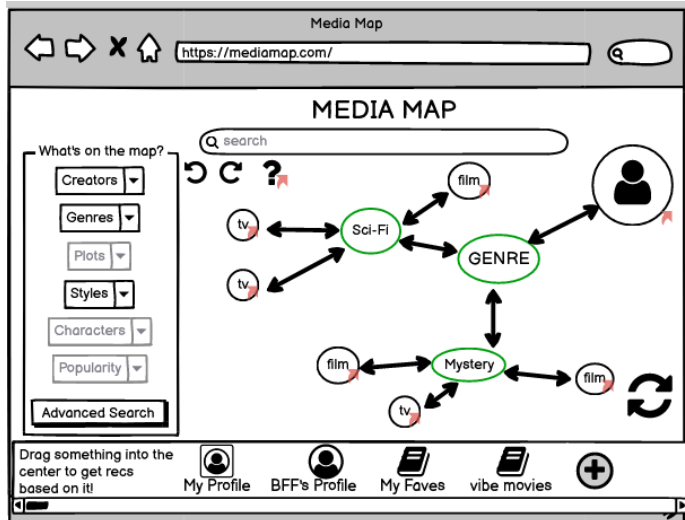
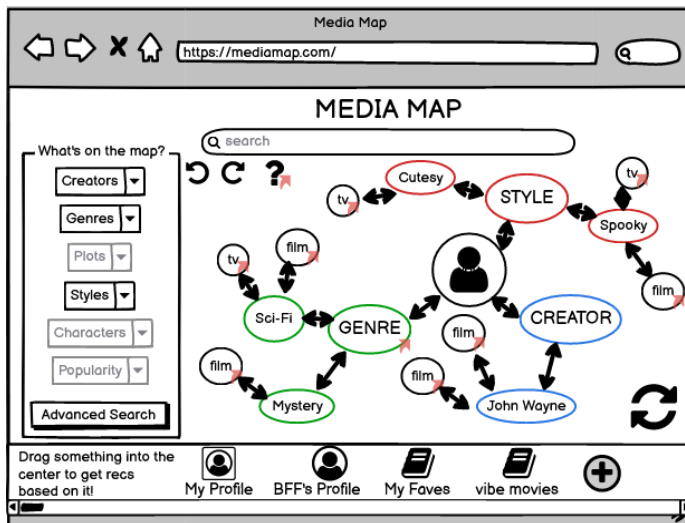


Nina is happy. Thanks to Media Map, she got to a TV show that she wanted.

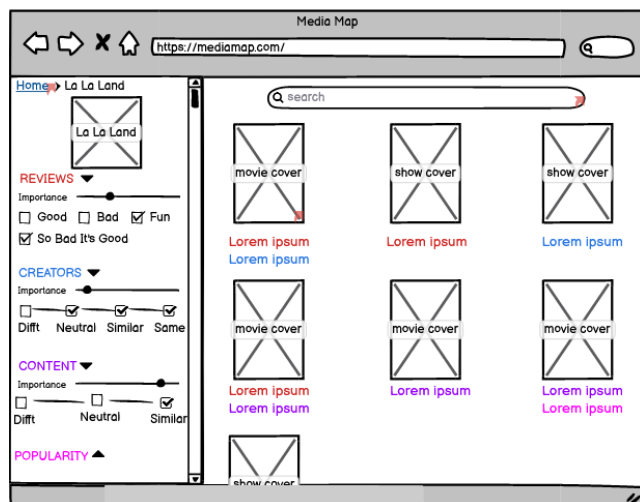
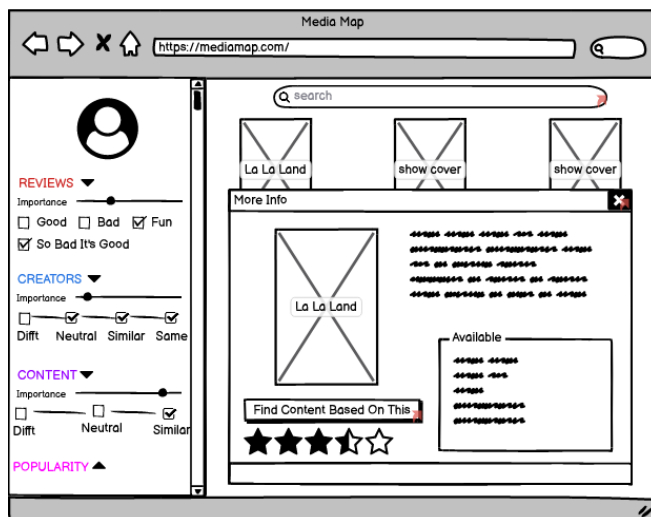
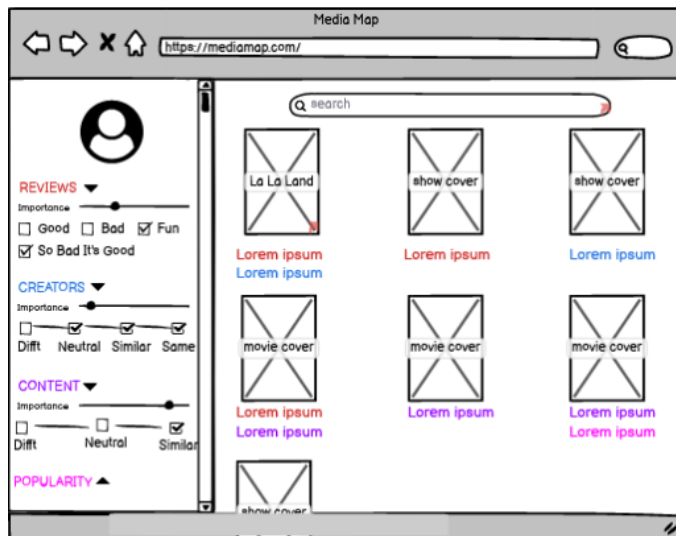
OPTION 1:



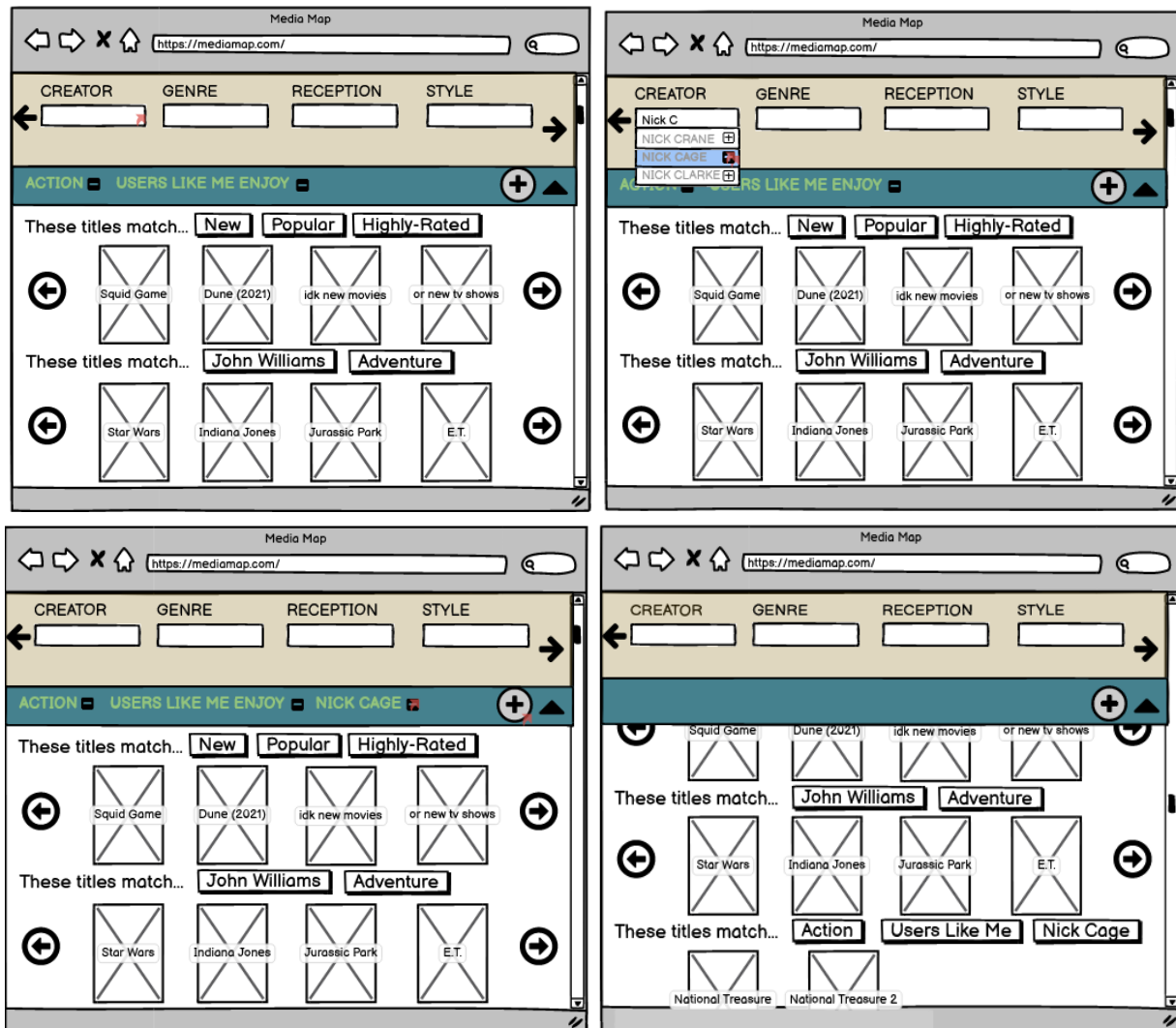
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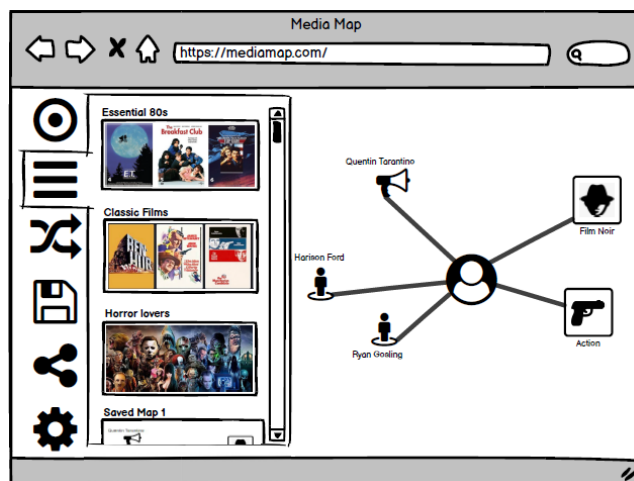
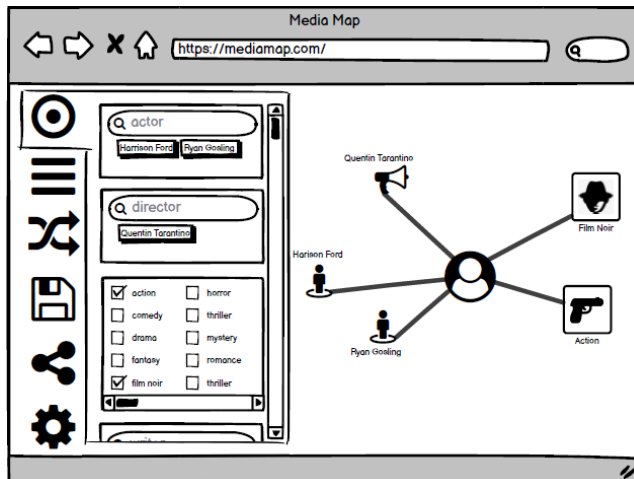
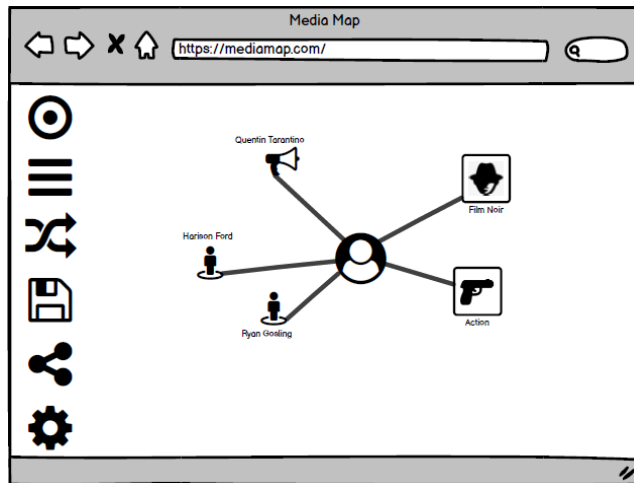
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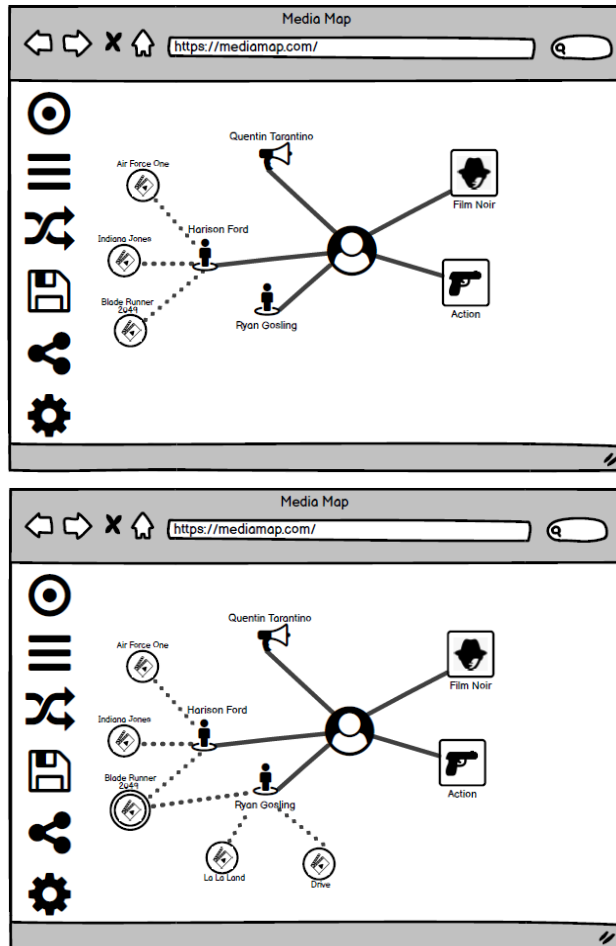


OPTION 4:



OPTION 5:





10. Risks to mitigate

From the Individual Project, we have reason to believe that a user need for our project exists, at least within the confined population of Columbia University students, and likely outside that niche as well. As such, our central risk is finding the best way to answer that need.

The first risk we wish to mitigate is that of the project's format. Our project should allow users to visualize, understand, and control connections between different movies and TV shows, but there are many ways of doing this—some better than others. By creating numerous designs and showing them to users, we will climb different design hills until we find the format that is best for users: the "right design".

Other design risks include those of usability. Once we have a format, how do we best utilize that format to make a finished project that is simple, elegant, and easy to use?

On the technical end, our risks are functional: will we be able to meet the standards set by the design? That is, generating the media map with simple and clear choices, such as "same actors" or "same director" is definitely feasible to implement. However, when it comes to the case that if users choose some abstract criteria, such as "similar stories," to generate the media map it could be harder to provide better recommendations than existing recommendation services like Netflix. On top of that, after generating a "media map" based on the user's choice of key attributes, we would like to show users which streaming services they can utilize to watch each movie and tv show on the map. However, as the streaming services update the list of movies and tv shows that are available based on regions, contracts,

and so on, we might not be able to reflect these updates accurately, which might end up providing outdated information.

11. Project tasks

A list of tasks that must be completed in order to finish the final project. This should include tasks that have been completed so far and tasks that are yet to be completed. For each incomplete task, state who is currently assigned to complete it (assigning multiple people is OK), how many hours you estimate the task will take to complete, and what your internal due date for completion is. For each complete task, state who completed it (stating multiple people is OK), how many hours it took to complete it, and the date that it was completed. Include ALL known tasks here, including implementation (coding), user testing, and project write up tasks.

TASKS COMPLETED

- Pick design topic (2 hrs, all, 11/13)
- Balsamiq prototypes for different formats (2 hrs, Adi, 11/15)
- Further developed ideas for different formats (1 hr, Adi and Swan, 11/16)
- Conduct research on which Web Frameworks / DBMS we will use for the project (2 hrs, Je Seung and Alexandra, 11/16)
- FP Progress Report I (all) 11/19
- Create low-fi prototypes for different formats (2 hrs, Adi + Swan). Internal due date: 11/19

TASKS INCOMPLETE

Project Write-up tasks

- First studio session (11/23, 2:40-2:55 PM)
- FP Progress Report II and 360° assessment (12/3, 3PM)
- final project turn-in (12/16, 11AM)
- final project video (12/15, 3PM)

Design tasks

- User interviews regarding different formats (30 min apiece, all). Internal due date: 11/29
- New design based on user interviews (1 hr) Internal due date: 12/3
- Low-fi prototype the new design (1 hr) Internal due date: 12/3
- User interviews regarding new design (30 min apiece, all) Internal due date: 12/7
- Update design based on secondary user interviews (1 hr) Internal due date: 12/10
- High-fi prototype the updated design (2 hrs) Internal due date: 12/10

Coding tasks

- Layout basic elements (forms, inputs, etc), including sign in and sign up page, profile pages, etc (10 hrs) Internal due date: 12/6 (consider extra work if the design changes)
- Create Sign in / Sign up APIs - integrate it with front end (5 hrs) Internal due date: 12/7
- Integrate Neo4j API with front end to make element interactive (5 hrs) Internal due date: 12/10
- Implement Neo4j with Flask to create a movie recommendation feature (10 hrs) Internal due date: 12/10
- Integrate both ends and test app (3 hrs) Internal due date: 12/13

TASK ASSIGNMENT

Specifics have yet to be decided, but generally, Adi and Swan will focus on the design tasks, and Alex and Jason will focus on the coding tasks.